

Information Retrieval is considered one of the most important fields in computer science because it focuses on storing, organizing, searching, and retrieving relevant information from massive collections of data efficiently and accurately. Modern search engines rely heavily on Information Retrieval techniques to analyze user queries, process keywords, and rank results according to their relevance. These systems use concepts such as indexing, tokenization, stemming, ranking algorithms, and query processing to help users find the most useful information in a very short time. With the rapid growth of digital data and the internet, the connection between Information Retrieval and Machine Learning has become extremely strong. Machine Learning enables systems to learn patterns from previous data and improve their performance automatically without being explicitly programmed for every situation. It is widely used in tasks such as text classification, recommendation systems, spam detection, sentiment analysis, and predictive analytics. Furthermore, advancements in Deep Learning and Neural Networks have significantly improved Natural Language Processing, allowing systems to better understand human language, context, and meaning rather than relying only on exact keyword matching. Today, these technologies are used in many real-world applications including web search engines, virtual assistants, healthcare systems, e-commerce platforms, social media analysis, cybersecurity, and big data analytics. The integration of Information Retrieval with Machine Learning has transformed the way intelligent systems interact with users and process information, making modern technology faster, smarter, and more capable of delivering accurate and personalized results.